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**Optimal Task Allocation using Modified
Primal Simplex for KNUST Cleaning Staff**

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Chapter 1

Introduction

1.1 Background of the Study

Task allocation as defined by [Sheridan \(2007\)](#) refers to the process of assigning various tasks to resources, tools, or agents capable of performing them, or conversely, allocating these resources to the tasks. This concept is central to workforce optimization, as highlighted by ([Mulla et al., 2016](#)), which underscores its critical role in helping organizations meet Service Level Agreements (SLAs) and achieve operational excellence. According to [H.A. Eiselta \(2008\)](#), effectively assigning tasks to employees is a challenging endeavor. Mistakes in this process can lead to significant issues, including decreased efficiency and deteriorating labor relations. Effective workforce planning is crucial for management in any organization, as it involves assigning employees with the right skills to appropriate positions at the optimal time. This planning challenge spans across multiple industries, including services, healthcare, education, military, and transportation, among others. For example, in the services industry, a study highlighted the difficulties associated with deploying service technicians to client locations, which led to increased costs and inefficiencies. To address these issues, researchers developed a workforce planning model aimed at reducing travel costs and improving operational efficiency according ([Yee et al., 2023](#)). Similar workforce planning challenges are evident in other sectors such as healthcare, education, military, and transportation. Given the broad and intangible nature of services, addressing these issues can be particularly complex and necessitates further research. Understanding and resolving these challenges can enhance operational efficiency and effectiveness across various industries. According to [Kabiru et al. \(2017\)](#) the assignment problem, a significant aspect of workforce planning, focuses on optimally

assigning tasks to individuals to minimize total costs. This problem is particularly relevant in resource-constrained environments where efficient task allocation is crucial. The assignment problem is characterized by the use of binary decision variables, ensuring that each task is assigned only once, with the requirement that functions must be linear or capable of being linearized. Various methods exist for solving these problems, with the "Hungarian method," developed by H.W. Kuhn in 1955, being one of the most effective. This method simplifies the task allocation process through matrix reduction, enabling the optimal assignment of tasks without the need for exhaustive comparisons. As a specialized form of the transportation problem, the assignment problem is fundamental in sectors where minimizing operational costs and time is critical, making it a vital area of study in workforce planning.

1.2 Problem Statement

The primary challenge identified in this study is the inefficient and uncoordinated task distribution among cleaners, where tasks are randomly assigned without considering effectiveness or time management, due to the absence of a strategic and optimized task allocation system.

1.3 Objectives

These goals have been set to help in designing and evaluating the task allocation model.

1. To formulate a model to optimize task allocation, ensuring efficiency and effective time management.
2. To develop a strategic task allocation plan for the cleaning staff using predefined information about the building and its operational requirements.
3. To conduct simulations with different scenarios to test the effectiveness of the proposed task allocation model.

1.4 Methodology

The total available time is segmented into time periods to structure the workflow for task completion. A linear programming model is developed for dynamic task assignment, which is then solved using the Primal Simplex Method.

1.5 Justification

This project aims to develop an optimized task allocation plan for cleaning staff in a building complex. The primary objective is to develop an algorithm that minimizes the total time required for all cleaning operations to be completed. By solving the model using primal simplex method, a task allocation plan is developed to ensure that all tasks are completed within the shortest possible time frame, thereby increasing overall productivity.

1.6 Case Study

The Aboagye Menyah Complex, located at Kwame Nkrumah University of Science and Technology, serves as our case study. This building complex has been selected to develop and test our task allocation plan for cleaning staff, ensuring that all cleaning tasks are managed efficiently and effectively within this environment.

1.7 Organization

This research is organized into five main sections. The first chapter, the Introduction, covers essential elements such as the study's background, objectives, and problem statement. Chapter two provides a thorough literature review, synthesizing and analyzing relevant studies by various authors in the field. The third chapter details the methodologies employed in the study, while the fourth chapter presents the results and discusses their implications. The final chapter offers a comprehensive conclusion and provides recommendations based on the study's findings.

Chapter 2

Literature Review

2.1 Introduction

Various studies and articles have explored methods for optimizing task allocation using algorithms like the Branch-and-bound, Greedy and Genetic algorithms. However, this study specifically focuses on developing an optimal task allocation plan for cleaning staff in building complexes using the Primal Simplex algorithm. Our approach aims to enhance efficiency, balance workloads, and minimize time spent on tasks by strategically assigning tasks based on predefined building requirements and operational conditions.

2.1.1 Definitions

1. **A building complex:** This is a collection of buildings designed and constructed concurrently on a single site in a common architectural theme or for a certain use. These structures can work together as a single unit or separately and can be vertically or horizontally connected. Residential complexes, business complexes, or institutional complexes like campuses for businesses, hospitals, or schools can be classified as building complexes.
2. **Task allocation:** [Sheridan \(2007\)](#) define task allocation to mean the assignment of various tasks that need to be done to resources, instruments, or agents capable of doing them or the allocation of such resources to tasks, which is equivalent.
3. **Task:** This refers to a specific cleaning activity assigned to a member of the cleaning staff within a building complex. This could include activities such as sweeping, mopping, dusting, trash removal, or any other cleaning duty required to maintain

the cleanliness and hygiene of the facility.

4. **Period:** A specific interval of time within which tasks are scheduled to be performed or completed. In our work, periods are used to organize and allocate tasks systematically, ensuring efficient management of the total available time for completing all tasks.

2.2 Task Allocation Problem

The problem in this study is a task allocation problem. Task allocation problems are a class of optimization problems in operations research and management science that involve assigning a set of tasks to a group of resources or agents. The goal is to allocate tasks in such a way that certain criteria, such as minimizing total completion time or maximizing efficiency, are optimized. Task allocation is the assignment of various tasks that need to be done to resources, instruments, or agents capable of doing them ([Sheridan, 2007](#)).

2.3 Task Allocation Methods

Task allocation methods are techniques or approaches used to assign tasks to resources or agents in an optimized manner. These methods aim to efficiently distribute tasks to achieve specific objectives, such as minimizing overall time, cost, or maximizing resource utilization. Some common task allocation methods include Integer Programming, Heuristic methods and Linear programming.

2.3.1 Integer Programming

Integer Programming (IP) is a specialized form of linear programming where some or all of the decision variables are restricted to integer values. This is particularly useful for problems where the decision variables represent discrete items or units, such as tasks that must be assigned to workers or machines.

2.3.2 Linear Programming

Linear Programming Problem, or LPP, is a mathematical method to determine the best possible outcome for a given set of parameters represented in linear relationships ([Dantzig, 1963](#)). It involves optimizing a linear objective function, subject to a set of linear constraints.

2.3.3 Heuristic method

Heuristic refers to experience-based techniques for problem solving, learning, and discovery. Where an exhaustive search is impractical, heuristic methods are used to speed up the process of finding a satisfactory solution. Examples of this method include using a rule of thumb, an educated guess, an intuitive judgment, or common sense ([Desale et al., 2015](#)).

2.4 Literature Review

Effective task allocation is critical in various operational settings, particularly when optimizing the performance of teams assigned to complex tasks. This challenge becomes even more pronounced when tasks require collaborative efforts from multiple agents, each bringing unique skills and capabilities to the table. As we delve into this subject, we encounter a range of methodologies designed to address these complexities. One such approach is detailed in a research work by ([Ellis, 2006](#)) on task assignment problems involving teams of agents, where the focus is on maximizing overall efficiency and effectiveness.

Many industries face significant challenges in workforce management, particularly in optimizing resource allocation to minimize costs and improve service efficiency. To address these issues, a study was recently conducted with the main aim of developing an integer programming (IP) model to optimize workforce planning in cleaning services, specifically targeting the reduction of hiring costs in a public university. The background of the study

highlights the lack of effective workforce planning in the industry, leading to inefficiencies such as high turnover rates and increased operational costs. This research was motivated by the need to enhance the performance of cleaning services operations by systematically addressing the allocation of resources and improving the overall efficiency of cleaning tasks within the constraints of the existing workforce. The proposed model evaluates various scenarios to provide actionable insights for improving workforce management in the cleaning industry (Yee et al., 2023).

Service delivery organizations face numerous challenges, ranging from operational inefficiencies to stringent service level agreements (SLAs) that must be met to maintain client satisfaction (Mulla et al., 2016). These challenges include managing high volumes of transactions, optimizing resource utilization, and ensuring fairness in task allocation. As organizations increasingly rely on advanced technology and automation, the complexity of these challenges grows. The consequences of failing to address these issues effectively can be significant, leading to missed deadlines, reduced productivity, and increased operational costs. Consequently, there is a growing need for sophisticated task allocation methods that can navigate these complexities and enhance overall performance, much like the approach taken in our study, where we focus on optimizing task distribution among cleaning staff to ensure efficient operation and resource management. Hence, in SDOs, the people involved in allocation of transactions need decision support system to help them. One of the main components of such a system is the task allocation tool developed from appropriate mathematical models and algorithms, which is the main focus of this work (Mulla et al., 2016).

In the work of H.A. Eiselta (2008) addressed the complex issue of task allocation and its impact on employee efficiency and job satisfaction. The study focused on minimizing inefficiencies caused by absenteeism and task misassignment by mapping employees and tasks in a skill space to optimize task allocation. The research incorporated various factors like employee boredom, workload equity, and costs into the task assignment process, aiming to enhance job satisfaction and reduce absenteeism. The study also examined

the correlation between job satisfaction and absenteeism, building on earlier theories and incorporating multidimensional skills and multiple objectives into their approach. This work diverged from traditional scheduling problems by focusing on long-term personnel retention through equitable workload distribution and task alignment with employee abilities.

In their research, ([Reinout Heijungs, 1998](#)) critically examined the allocation problem in life cycle inventory analysis, arguing that the common definition of the problem where a unit process produces more than one function is insufficient. They proposed a more precise and operational definition, allowing for a systematic categorization of approaches to address the allocation problem. Through hypothetical cases, they illustrated that not all multifunctional processes, such as those involving internal (closed-loop) recycling, necessarily present an allocation issue. Their work highlighted the need for a deeper understanding of the allocation problem beyond traditional approaches.

An article by ([Cheng et al., 2018](#)) aimed to enhance the efficiency and flexibility of manufacturing systems by reviewing and optimizing task allocation processes. The study was motivated by the need to adapt to the evolving demands of modern manufacturing environments, where traditional system-oriented decision-making methods often fall short. The authors provided a comprehensive background on task allocation, highlighting its critical role in ensuring that resources are effectively utilized. They categorized the task allocation process into six key stages: task description and modeling, analysis and modeling of the task allocation process, algorithm selection, decision-making, simulation, and task execution. The study emphasized the importance of integrating intelligent systems and advanced algorithms to improve task allocation outcomes. The authors also identified gaps in current research and proposed future directions to further advance intelligent task allocation in manufacturing, aiming to address the limitations of existing approaches and better meet the needs of dynamic manufacturing systems.

Task allocation and load balancing are critical challenges in distributed systems, where

multiple computational units are connected through networks to perform large-scale, high-performance tasks. As distributed systems become more complex and diverse, optimizing the allocation of tasks and balancing the load across various nodes is essential to ensure efficient operation, reliability, and scalability. This study aims to provide a comprehensive review of existing research on task allocation and load balancing across different types of distributed systems, such as grids, peer-to-peer (P2P) systems, ad hoc networks, and cloud computing systems. Unlike previous surveys that focused on a single type of system, this research categorizes and examines the common characteristics and challenges shared by various distributed systems. The motivation behind this study stems from the need to address the complexities arising from the lack of a global control unit, the heterogeneous nature of nodes, the distribution and sharing of resources, and the constraints imposed by network structures. By reviewing control models, resource optimization methods, reliability mechanisms, coordination strategies, and the impact of network structures on task performance, the study offers a macroscopic overview of the field, helping to correlate findings across different systems and suggesting directions for future research ([Yichuan Jiang, 2016](#)).

The assignment problem, a key area of linear programming (LP), focuses on assigning individuals to tasks to minimize total costs. This problem is characterized by the use of binary decision variables to indicate task completion, the rule that each task can only be assigned once, and the need for functions to be linear or linearizable. LP, as defined by Lomba (1964), is a mathematical method for determining optimal task assignments given resource constraints, using structured mathematical relationships (Williams, 1995). Among the various methods for solving assignment problems, the "Hungarian method," developed by H.W. Kuhn in 1955, is widely recognized for its efficiency. This method uses matrix reduction to simplify the problem, allowing for the optimal assignment without direct comparison of all options ([Kabiru et al., 2017](#)).

An article by ([Junyi Zhang, 2005](#)) developed a household task allocation and time use model that incorporates group decision-making dynamics. The study sought to over-

come the limitations of traditional activity-based models, which often focus on individual decision-making, by factoring in the collective influence of all household members on task distribution. Utilizing a multi-linear group utility function, the model was validated with activity-travel diary data. The findings underscored the importance of travel time in enhancing the model's accuracy, particularly for predicting joint and assigned activities. Additionally, the study revealed that in nearly half of the households, the husband exerted the most influence on decision-making. The authors suggested expanding the model to include more activities, additional household members, and other decision-making aspects such as mode choice and timing, offering a roadmap for future research.

2.5 Summary

Based on the review of various articles, we found that effective task allocation strategies significantly enhance the efficiency of task distribution. These publications presented numerous approaches and models for optimizing task allocation. Following our evaluation, we implemented a linear programming model to develop an optimized task allocation plan for managing cleaning staff in building complexes.

Chapter 3

Methodology

3.1 Introduction

This chapter outlines the process used to create an optimized task allocation plan utilizing linear programming for a building complex. As part of our methodology, an optimization model is formulated to assign a number of tasks to cleaning staff in a building complex. The model is solved using the Primal Simplex Method. The main processes in our methodology are described in this chapter, starting with creating the framework for task allocation within the Aboagye Menyah complex and ending with the formulation and solution of the optimization model.

3.2 Assumptions

To solve our problem, we made the following assumptions:

1. There is a perfect efficiency in collaboration among workers.
2. The cleaning staff possess uniform skills and abilities, making them equally capable of completing assigned tasks in the same amount of time.

3.3 Model Description

3.3.1 Parameters

1. **T** - Total Available time in minutes.

2. $\mathbf{T}_j$ - Average time taken to for one worker to complete task j .
3. $\mathbf{W}$ - Total number of workers.
4. $\mathbf{N}$ - Total number of Task to complete.
5. $\mathbf{M}$ - Maximum allowable task completion time per worker.
6. $\mathbf{P}$ - Maximum number of periods.
7. $\mathbf{L}$ - Maximum time spent in any period.

3.3.2 Decision Variables

1. $x_{ijt} \in \{0, 1\}$

$$x_{ijt} = \begin{cases} 1 & \text{if worker } i \text{ is assigned to task } j \text{ in period } t \\ 0 & \text{Otherwise} \end{cases}$$

2. $p_t \in \{0, 1\}$

$$p_t = \begin{cases} 1 & \text{if period } t \text{ is active} \\ 0 & \text{Otherwise} \end{cases}$$

3. $tp_{j,t} \in \{0, 1\}$

$$tp_{j,t} = \begin{cases} 1 & \text{if task } j \text{ is assigned in period } t \\ 0 & \text{Otherwise} \end{cases}$$

4. $L \in \mathbb{R}$

Maximum time spent in any period

5. $y_{j,t} \in \mathbb{R}$

Time taken to complete task j in period t

3.3.3 Task Time and Data Extraction

In cleaning operations, estimating task times accurately is crucial for effective workforce planning. Traditional models often overlook the impact of fatigue on task performance. As cleaners work for extended periods, their productivity can decrease due to fatigue, leading to longer times required to complete cleaning tasks. To address this issue, we have developed a function that incorporates fatigue into task time estimates, enhancing the accuracy and practicality of our model.

Task Time Function Overview

The fatigue adjustment function modifies task time estimates to reflect the increasing time required as cleaners become more fatigued. The function takes into account the number of tasks completed and the duration of the cleaning session to adjust the average time needed for subsequent tasks.

Mathematical Formulation

The function is represented by the following equation The time to clean the n -th tile:

$$T(n) = T_{\text{base}} \cdot e^{\alpha n} \quad (3.1)$$

where $T(n)$ is the time to clean the n -th tile, α controls how quickly the fatigue sets in and T_{base} is the initial time estimate for each task before considering fatigue.

The total time to clean all tiles up to N tiles is the sum of the times to clean each tile

$$T_{\text{total}} = \sum_{n=1}^N T_{\text{base}} \cdot e^{\alpha n} \quad (3.2)$$

This is a geometric series and can be simplified using the formula for the sum of a geometric series

$$\sum_{n=1}^N e^{\alpha n} = \frac{e^{\alpha}(1 - e^{\alpha N})}{1 - e^{\alpha}} \quad (3.3)$$

Therefore, the total cleaning time becomes

$$T_{\text{total}} = T_{\text{base}} \cdot \frac{e^{\alpha}(1 - e^{\alpha N})}{1 - e^{\alpha}} \quad (3.4)$$

Parameter Definitions:

T_{total} *Total time*: The total estimated time required to clean all tiles (WCs or sinks).

n *Task index*: The index of the tiles (WCs or sinks) in the sequence from 1 to N .

N *Number of tasks*: The total number of tasks to be completed i.e.(Total number of tiles or WCs or sinks).

3.4 Model Formulation

3.4.1 Objective Function

Minimize L

3.4.2 Constraints

1.

$$y_{j,t} \leq \frac{T_j}{\sum_{i=1}^W x_{i,j,t}}, \quad \forall j, t. \quad (3.5)$$

This constraint ensures that when multiple workers are assigned to a task, the task's duration is proportionally reduced based on the number of workers assigned.

2.

$$\sum_{t=1}^P tp_{j,t} = 1 \quad \forall j \quad (3.6)$$

This constraint ensures that each task is assigned to exactly one period.

3.

$$\sum_{j=1}^N x_{ijt} = p_t \quad \forall i, t. \quad (3.7)$$

This constraint ensure that every worker is assigned only one task in any active period and also ensures that every is working in any active period.

4.

$$p_t \geq x_{ijt}, \quad \forall i, j, t. \quad (3.8)$$

This constraint ensure that period t is active if a worker is assigned in that period.

5.

$$L \geq \sum_{j=1}^N y_{j,t} \quad \forall t \quad (3.9)$$

The constraint ensures that L reflects the maximum time spent in any single period among all periods considered (P periods).

6.

$$\sum_{i=1}^W \sum_{t=1}^P x_{i,j,t} = 1 \quad \forall j \text{ where } T_j < M \quad (3.10)$$

$$\sum_{i=1}^W \sum_{t=1}^P x_{i,j,t} \geq 1 \quad \forall j \text{ where } T_j \geq M \quad (3.11)$$

Each task with completion time greater than M should be assigned to more than one worker

3.5 Simplex Method

The simplex algorithm is the most used tool for solving linear programs. It is an iterative method that was developed by (Dantzig, 1963).

An initial feasible basis constitutes a necessary element for starting the simplex algorithm. The two-phase and the big-M methods are the most familiar techniques for the research of an initial feasible basis. Each of these methods brings in artificial variables, which lead in consequence to a growth in the size of the initial standard form (Nabli, 2009). This method easily provides sensitivity analysis, which is critical for understanding the impact of parameter changes. The Simplex method's use of explicit bases and direct computation of dual variables makes it well-suited for sensitivity analysis, providing clear and interpretable information about the impact of changes in the model parameters (Wright, 1997). Also according to (Vanderbei, 2001), the simplex method consistently performs well across a wide range of LP problems. Numerous empirical studies and practical applications have shown that the Simplex method solves LP problems efficiently and effectively in real-world scenarios.

3.5.1 Primal Simplex Method

The Primal Simplex Method is one of several variations of the simplex method used in linear programming. The Primal Simplex method starts with a feasible solution and iterates to improve the objective function while maintaining feasibility at every step. It moves from one basic feasible solution (BFS) to another, ensuring that the constraints are always satisfied. It operates directly on the primal form of the linear programming problem. In contrast, the Dual Simplex method operates on the dual of the linear program, and the Revised Simplex method uses a more computationally efficient version of the simplex method by keeping only necessary parts of the tableau.

Algorithm

PRIMAL SIMPLEX ([Richard V. Helgason, 1995](#))

Inputs:

A - $m \times n$ constraint matrix

c - m -component vector of unit costs

u - n -component vector of upper bounds

b - m -component right-hand-side vector

Output:

(x_B, x_N) , an optimal basic feasible solution for $\min\{cx : x \in X\}$

Assumptions:

1. $\text{rank}(A) = m$
2. $X = \{x : Ax = b, 0 \leq x \leq u\} \neq \emptyset$

begin

[Initialization] $\text{optimal} \leftarrow \text{no}$;

let $(x_B, x_N) \in X$ be any basic feasible solution with A , c , and u partitioned in corresponding fashion as $[B \mid N]$, $(c_B \mid c_N)$, and $(u_B \mid u_N)$;

while $\text{optimal} = \text{no}$ **do**

 [**Dual Calculation**] $\pi \leftarrow c_B B^{-1}$

 [**Pricing**] $L \leftarrow \{j : \pi N_j - c_N > 0 \text{ and } x_N = 0\}$;

$M \leftarrow \{j : \pi N_j - c_N < 0 \text{ and } x_N = u_N\}$;

if $L \cup M = \emptyset$ **then**

optimal $\leftarrow$ yes

else

select $k \in L \cup M$

[Column Update] $y \leftarrow B^{-1}N_k$

[Ratio Test] $R \leftarrow \{i : y_i > 0\}$

$S \leftarrow \{i : y_i < 0\}$

if $k \in L$ then

$$\Delta \leftarrow \min \left\{ u_{N_k}, \min_{i \in R} \frac{x_{B_i}}{y_i}, \min_{i \in S} \frac{(u_{B_i} - x_{B_i})}{-y_i} \right\}$$

$$T \leftarrow \{i \in R : \frac{x_{B_i}}{y_i} = \Delta\} \cup \{i \in S : \frac{(u_{B_i} - x_{B_i})}{-y_i} = \Delta\}$$

else

$$\Delta \leftarrow \min \left\{ u_{N_k}, \min_{i \in R} \frac{(u_{B_i} - x_{B_i})}{y_i}, \min_{i \in S} \frac{x_{B_i}}{-y_i} \right\}$$

$$T \leftarrow \{i \in R : \frac{(u_{B_i} - x_{B_i})}{y_i} = \Delta\} \cup \{i \in S : \frac{x_{B_i}}{-y_i} = \Delta\}$$

[Value Update]

if $k \in L$ then

$$x_{N_k} \leftarrow \Delta \text{ and } x_B \leftarrow x_B - \Delta y$$

else

$$x_{N_k} \leftarrow u_{N_k} - \Delta \text{ and } x_B \leftarrow x_B + \Delta y$$

if $\Delta \neq u_{N_k}$ then

[Basis Exchange Update]

select $j \in T$

interchange x_{B_j} with x_{N_k} to form a new partitioning $[B \mid N]$

endif

endif

endwhile

end

Chapter 4

Results and Discussions

4.1 Introduction

In this chapter, we present the results of our task allocation plan for cleaning staff in a building complex. Using the method outlined in the previous chapter, we provide the data, analyses, and results in tables to demonstrate the effectiveness of our approach in optimizing task allocation and ensuring efficient use of resources.

4.2 Results and Analysis

The model was ran using data from our case study building, which included a total of 12 workers. The tasks were subjected to a completion schedule, with a maximum of 10 periods allocated for task execution. The maximum length of time that a worker can be assigned to tasks within a single period, denoted by M , is 40 minutes. These parameters were crucial in evaluating the effectiveness of the task allocation plan.

4.2.1 Data Extraction from Case Study Building

The table below presents the number of tiles, sinks, and WCs in the rooms designated for cleaning within the Aboagye Menyah Complex.

Table 4.1: Room Specifications for Cleaning at Aboagye Menyah Complex

Room Description	Tile Type	Number of Tiles	Number of WCs	Number of Sinks
SF1	Small	556	0	0

Room Description	Tile Type	Number of Tiles	Number of WCs	Number of Sinks
FF1	Small	558	0	0
TF1	Small	556	0	0
GF1	Small	596	0	0
SF7	Small	548	0	0
FF5	Small	548	0	0
FF6	Small	515	0	0
SF8	Small	515	0	0
SF19	Large	80	0	0
FF24	Small	570	0	0
FF2	Medium	111	0	0
SF20	Large	80	0	0
TF34	Small	515	0	0
GF17	Small	531	0	0
GF18	Small	369	0	0
GF19	Small	255	0	0
STWCGF	Large	99	7	3
STWCFF	Large	99	8	3
STWCSF	Large	99	8	3
TWCTF	Large	99	8	3
STWCB	Small	39	4	2
TWCB	Small	60	4	6
TWCG	Small	60	4	6
TWCG	Small	60	4	6
TWCG	Small	60	4	6
TWCG	Small	60	4	6
TWCF	Small	60	4	6
TWCF	Small	60	4	6

Room Description	Tile Type	Number of Tiles	Number of WCs	Number of Sinks
TWCF	Small	60	4	6
TWCF	Small	60	4	6
TWCS	Small	60	4	6
TWCS	Small	60	4	6
TWCS	Small	60	4	6
TWCS	Small	60	4	6
TWCT	Small	60	4	6
TWCT	Small	60	4	6
TWCT	Small	60	4	6
TWCT	Small	60	4	6

Equation 3.4 is used to derive the table below detailing the derived task times associated with each room using the data in table 3.1.

Table 4.2: Task Time Assignments

Room Description	Task Time (mins)
SF1	87
FF1	88
TF1	87
GF1	106
SF7	84
FF5	84
FF6	71
SF8	71
SF19	13
FF24	94
FF2	12

Room Description	Task Time (mins)
SF20	13
TF34	71
GF17	77
GF18	33
GF19	17
STWCGF	42
STWCFF	44
STWCSF	44
TWCTF	44
STWCB	21
TWCB	31
TWCG	31
TWCG	31
TWCG	31
TWCG	31
TWCF	31
TWCF	31
TWCF	31
TWCF	31
TWCF	31
TWCS	31
TWCS	31
TWCS	31
TWCS	31
TWCT	31
TWCT	31
TWCT	31
TWCT	31

4.2.2 Period Assignments

Period	Time Frame	Duration (minutes)
Period 1	6:00 AM - 6:36 AM	36.00
Period 2	6:41 AM - 7:13 AM	32.00
Period 3	7:18 AM - 7:57 AM	39.00
Period 4	8:02 AM - 8:33 AM	31.00
Period 5	8:38 AM - 9:14 AM	36.00
Total	—	174.00

Table 4.3: Period Assignments

4.2.3 Tasks Assignments

Period 1

Task ID	Task Type	Assigned Workers
Task 4	GF1	Worker 2, Worker 8, Worker 9
Task 8	SF8	Worker 3, Worker 12
Task 15	GF18	Worker 11
Task 16	GF19	Worker 7
Task 18	STWCFF	Worker 5, Worker 6
Task 23	TWCG	Worker 4
Task 28	TWCF	Worker 10
Task 38	TWCT	Worker 1

Table 4.4: Task Allocation for Period 1 (36.00 minutes)

The scenarios presented above outline the task allocation within the Aboagye Menyeh Complex, demonstrating how tasks are optimally assigned to workers under various constraints. Table 4.4 shows the optimized task allocation in Period 1. Tables ??, ??, and ?? detail the task allocations in Periods 2, 3, and 4, respectively. The task allocations in Period 5 is shown in Tables ?? highlighting the efficiency of the optimized allocation plan.

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

In conclusion, a strategic task allocation plan was developed for the cleaning staff at the Aboagye Menyeh Complex. A linear programming model was formulated to optimize the allocation of tasks, ensuring efficient use of resources and adherence to time constraints. The model was tested using case study data, and it successfully achieved optimal results, demonstrating its effectiveness in enhancing task management within the building complex.

5.2 Recommendation

It is recommended that the strategic task allocation plan and linear programming model developed in this study be implemented and regularly updated to reflect any changes in the building's cleaning requirements or staff availability. Additionally, integrating this model with real-time data could further enhance its effectiveness, allowing for dynamic adjustments to task allocations based on actual conditions, such as unexpected staff absences or changes in cleaning priorities. This approach will ensure that the cleaning operations remain efficient and responsive to any operational challenges.

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